

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Thermal & oil pollution - Set 1

Mark scheme

AQA spec 3.4.3.2.6-3.4.3.2.7; Pollution; textbook pp. 254-262

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• As temperature increases the maximum dissolved oxygen concentration decreases (non-linear: falls faster at low temperatures)	1
01.2	• About 39 °C (accept 38-40)	1
01.3	• 9.1 - 7.9 (value at 27 °C, accept 7.8-8.0) • = 1.2 mg dm ⁻³ (accept 1.1-1.3)	2
01.4	• Figure 1 shows the maximum / saturation values; processes that remove oxygen (aerobic decomposition of organic matter / respiration) can give values below the line • Higher temperature increases the metabolic rate of organisms and the rate of bacterial decomposition so oxygen is used faster; values above the line are impossible	2
01.5	<i>Any 3 from:</i> • Increased metabolic rate so more energy used / less surplus for growth • Eggs develop faster / hatch earlier (out of step with food supply) • Non-indigenous warm-water species may thrive and out-compete indigenous species • Toxicity of other pollutants is greater as toxins are metabolised faster • Reduced resistance to disease • Proteins / enzymes denatured if temperature is very high (above about 40 °C)	3
01.6	• Hot water is sprayed from a shower floor in a tall hollow tower; air absorbs the heat and rises, drawing in more air (updraught), so the water returned is cooler (and saturated with oxygen)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 256-257; AQA spec 3.4.3.2.6.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from:</i> • Discharge of tank-washing water from tankers • Oil rig / drilling accidents / leakage during drilling • Pipeline leaks • Leakage from storage tanks / refinery spills • Waste lubricating oil from vehicles / industry washed into drains • Other ship accidents (fuel oil)	3
02.2	• Oil sticks feathers together so they lose their insulating properties and birds die of hypothermia • Feathers lose buoyancy / waterproofing so heavily oiled birds drown • Birds swallow toxic oil when preening; time spent cleaning feathers reduces time for feeding themselves / their young	3
02.3	<i>Any 2 from:</i> • The surface film reduces the diffusion of oxygen from the atmosphere into the water, so aerobic organisms suffer from low dissolved oxygen • Toxic components (benzene, xylene) dissolve and poison plankton near the surface • Oil blocks light so algae / phytoplankton cannot photosynthesise • Oil masks odours / narcotises so animals cannot find mates or food by smell • Shore organisms (molluscs, crustaceans) are covered and asphyxiated	2
02.4	• Dispersal / dilution by currents and waves is prevented so oil stays concentrated; in open water most spills disperse in days or weeks • Inshore areas have more sensitive ecosystems (shores, reefs, mangroves, breeding sites) and more human activities (fishing, tourism) to be affected	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-259; AQA spec 3.4.3.2.7.	
03.1	• Double hull: a gap of up to 2 m between the hulls so damage to the outer hull in a collision / grounding does not release oil • Inert gas system: cooled engine exhaust gas (no oxygen) fills the space above the oil so an explosive oil vapour / air mixture cannot form • Separate ballast tanks: ballast water is not carried in oil tanks so no residual oil is discharged when ballast is pumped out • Twin engines / rudders: if one fails the ship can still be steered / propelled and is not carried onto rocks by currents or wind • Crude oil washing: sludges are mixed and removed with the cargo so tanks do not need washing with seawater that was then discharged; oily water is unloaded at the terminal	5
03.2	• Offshore routes keep tankers further from the coast so if there is engine failure there is more time to deal with it before the ship is carried ashore • GPS gives accurate navigation; AIS lets ship positions be monitored and coordinated to avoid collisions and groundings	2
03.3	• The tank stands on an impermeable base surrounded by a wall • The volume enclosed by the wall is greater than the tank volume so all the oil is contained if the tank splits / cannot overflow	2
03.4	• Separates floating oil from drainage water so the oil is retained (for later treatment) and only water flows to the drains / river	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 259-260; AQA spec 3.4.3.2.7.	

Question	Marking guidance	Marks
04.1	- Booms are floating tubes with a skirt that restrict the movement of floating oil, keeping it near the source / away from sensitive sites (mangroves, reef, marine park) - They work best in sheltered / calm water such as the lagoon - Limitation: they do not work in open water where currents or waves are strong; oil can flow under the skirt in a strong current	3
04.2	- Skimmers: rotating (metal) discs / belts pick up floating oil which is scraped off and stored - Absorbent materials (high surface area, eg textile mops, straw, human hair) soak up oil and are then removed - Collected oil / absorbents are incinerated or sent to landfill (or oil recycled)	3
04.3	- Dispersants break the oil into an emulsion of droplets, making it more mobile so it spreads through the water column (to the reef / seabed organisms) rather than staying on the surface - They do not reduce the amount of oil; the dispersant / oil mixture may be toxic to corals and plankton in shallow, enclosed water	2
04.4	- Bacteria that break down hydrocarbons digest the residual oil (most rapid where temperature, oxygen and nutrients are high) - Accelerated by adding nutrients (nitrates / phosphates) and aerating (ploughing) / irrigating the sediment	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 261-262; AQA spec 3.4.3.2.7.	
05.1	$(2900 - 1400) / 1400 \times 100$ $= 107\%$ (accept 107.1)	2
05.2	- Large decrease from 24 per year (1975-79) to 2 per year (2010-14) - Steepest fall between the late 1970s and 1980s; slower decline / levelling off after 1995	2
05.3	<i>Any 3 from:</i> - Double-hulled tankers required for all large tankers - Improved navigation (GPS, AIS) and offshore shipping routes - Inert gas systems reducing explosions - Twin engines / rudders reducing groundings after mechanical failure - Stricter regulation, inspection and polluter-pays liability (eg after Exxon Valdez) - Separate ballast tanks / crude oil washing	3
05.4	- Most oil pollution comes from dispersed / diffuse sources not tanker accidents: waste lubricating oil, industrial oil in drains, road runoff - Routine operations: small spills, pipeline leaks, storage tank leakage, oil rig accidents (eg Deepwater Horizon) - Growth in oil use and transport means more small releases even if large accidents are rarer	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 258-262; AQA spec 3.4.3.2.7.	
06.1	B - Steam washing (hot water / steam jets kill organisms in the sediment)	1
06.2	- Chemicals cause the oil molecules to join together / solidify - Producing more solid material that can be collected / lifted more easily than liquid oil	2

Question	Marking guidance	Marks
06.3	• Air contains about 210 cm³ of oxygen per litre whereas water at 20 °C holds a maximum of about 6 cm³ per litre (less than 3% of the amount in air) • So a small amount of oxygen-consuming activity (decomposition, respiration) or warming can remove most of the oxygen; many aquatic organisms are near the lower end of their range of tolerance	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 254-262; AQA spec 3.4.3.2.6-3.4.3.2.7.	